

HAKODATE, Japan

WMO: 474300

Lat: 41.8167N

Lon: 140.7542E

Elev: 144

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.5	17.0	6.7	7.8	18.7	9.1	8.8	20.7	25.5	32.8	23.1	30.7	6.6	310	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.2	83.2	73.8	80.8	72.5	78.6	71.2	75.5	80.9	74.2	79.2	72.8	77.3	9.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	126.7	78.7	72.5	121.1	77.8	71.1	115.3	76.7	39.2	81.3	38.0	79.0	36.7	77.6	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	20.7	18.4	DB	10.4	86.7	3.2	2.9	8.1	88.9	6.3	90.6	4.5	92.2	2.1	94.3
			WB	9.3	76.8	3.1	1.6	7.1	78.0	5.3	78.9	3.5	79.9	1.3	81.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	27.1	28.4	35.0	45.1	54.3	61.4	69.2	72.3	66.1	54.5	42.5	31.6
	DBStd	16.52	5.40	5.87	5.76	5.38	4.88	4.59	4.85	4.49	5.12	5.75	7.14	6.23
	HDD50	2793	711	604	464	164	16	0	0	0	24	240	570	
	HDD65	6282	1176	1024	928	596	332	124	17	3	46	326	676	1034
	CDD50	2459	0	0	0	18	150	343	596	691	482	164	15	0
	CDD65	474	0	0	0	0	1	17	148	229	78	1	0	0
	CDH74	2004	0	0	0	0	6	39	519	1161	278	1	0	0
	CDH80	284	0	0	0	0	0	1	58	195	30	0	0	0
	WSAvg	8.0	8.4	8.6	9.3	8.9	7.9	7.1	6.8	7.0	7.3	8.0	8.6	8.7
	PrecAvg	47.0	3.1	2.6	2.5	2.8	3.7	3.3	4.9	6.1	5.8	4.1	4.4	3.7
(15) Precipitation	PrecMax	62.2	7.8	8.0	4.4	5.3	6.2	7.2	13.1	14.3	10.8	7.8	8.7	6.8
	PrecMin	34.7	1.7	0.9	0.7	0.3	0.8	1.1	1.1	0.9	1.3	1.3	2.1	1.8
	PrecStd	6.3	1.3	1.5	0.9	1.3	1.3	1.9	2.5	3.1	2.4	1.8	1.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.9	47.5	54.7	66.5	74.1	78.0	84.5	87.2	83.5	71.9	63.7	52.8	
	0.4% MCWB	39.9	42.6	46.8	53.8	58.9	65.2	74.6	75.8	73.8	62.9	56.0	48.6	
	2% DB	40.5	43.1	50.8	61.5	69.6	74.5	81.5	84.4	79.7	69.0	60.0	48.4	
	2% MCWB	36.6	38.1	44.2	50.8	57.2	63.7	72.6	74.8	71.4	60.9	53.9	44.6	
	5% DB	37.9	40.3	47.7	57.7	66.4	71.8	79.1	82.1	77.0	66.7	57.1	44.6	
	5% MCWB	34.2	36.0	41.8	48.8	56.0	63.1	71.6	73.4	68.8	59.2	51.6	40.5	
	10% DB	35.7	38.0	44.8	54.7	63.4	69.5	76.8	79.9	74.8	64.6	54.4	41.5	
	10% MCWB	32.1	33.9	39.4	47.2	54.7	62.3	70.5	72.4	67.4	57.9	49.5	37.6	
	0.4% WB	40.9	43.5	48.5	56.0	61.9	68.2	75.9	77.8	75.4	66.1	57.9	49.2	
	0.4% MCDB	42.8	46.6	52.3	61.6	69.4	73.3	81.5	83.5	81.1	69.9	61.1	51.8	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	37.3	39.1	45.5	53.0	59.6	66.2	74.5	76.3	73.4	62.9	55.2	45.0	
	2% MCDB	39.9	42.2	49.9	58.6	66.3	71.3	79.4	81.9	77.9	66.8	59.1	48.2	
	5% WB	34.8	36.6	42.6	50.5	57.7	64.6	73.2	75.1	71.3	60.9	52.4	41.0	
	5% MCDB	37.3	39.8	46.3	56.0	64.1	69.8	77.7	80.4	75.3	65.3	56.4	43.9	
	10% WB	32.5	34.2	40.0	48.3	56.0	63.1	71.6	73.8	68.9	58.9	49.7	37.8	
	10% MCDB	35.4	37.5	44.2	53.7	61.9	68.2	75.8	78.6	73.2	63.8	53.9	41.2	
(35) Mean Daily Temperature Range	MDBR	11.2	12.1	12.9	14.7	14.4	12.5	10.7	11.2	13.9	15.4	13.3	10.9	
	5% DB MCDBR	12.5	13.8	16.5	19.1	19.9	16.5	13.4	13.8	15.1	16.3	17.3	14.4	
	5% DB MCWBR	10.7	11.3	12.1	11.7	10.8	8.7	6.7	6.9	8.6	10.4	13.4	12.6	
	5% WB MCDBR	12.5	13.9	15.5	16.8	16.6	13.6	11.4	11.8	12.6	14.2	16.5	14.2	
	5% WB MCWBR	11.6	12.4	13.0	11.7	10.1	7.9	6.1	6.6	8.3	10.6	14.3	13.5	
(40) Clear-Sky Solar Irradiance	taub	0.355	0.401	0.473	0.529	0.546	0.518	0.514	0.482	0.424	0.407	0.392	0.371	
	taud	2.116	1.991	1.880	1.806	1.853	2.008	2.092	2.199	2.316	2.221	2.154	2.101	
	Ebn at Noon	239	245	241	238	238	245	245	249	256	243	224	219	
	Edh at Noon	34	45	56	65	64	55	50	44	36	35	32	32	
(44) All-Sky Solar Radiation	RadAvg	420	651	1038	1418	1585	1553	1362	1328	1169	872	506	355	
	RadStd	26	59	72	148	145	137	132	122	109	62	47	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.22	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page